Curated Research Articles

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- **Amorphous Sulfo-Halide Solid Electrolytes With Enhanced Anion Dynamics for Highly Stable All-Solid-State Sodium Batteries** — score: 1.000 This study presents a novel amorphous sulfo-halide solid electrolyte that enhances ionic conductivity and mechanical flexibility for all-solid-state sodium batteries. By integrating highly polarizable $S2-$ anions into a $TaCl5$ framework, the electrolyte shows significant improvements in ionic transport and structural stability, leading to high coulombic efficiency and substantial capacity retention over extensive cycling. These results suggest a promising approach for designing advanced sodium-ion solid-state electrolytes. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Elastomeric Electrolyte With Selective Tuning of Ion Transport Pathway for Low-Temperature and High-Voltage Lithium Metal Batteries** — score: 0.800 The study presents an innovative elastomeric electrolyte that improves the performance of lithium metal batteries at low temperatures by optimizing ion transport pathways within a bicontinuous structure of a polymer matrix and plastic crystals. This design not only enhances ionic conductivity and $Li+$ transference but also maintains mechanical integrity, resulting in exceptional cycling performance and capacity retention at low temperatures and high voltages. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **A balanced approach to oxygen electrodes** — score: 0.400 The article discusses advancements in reversible solid oxide cells, highlighting a novel design that enhances the performance and stability of oxygen electrodes by improving the balanced transport of oxide ions and electrons, thereby facilitating faster oxygen exchange. Journal: *Nature Energy*
- **Alternative pulse current discharge strengthens solid electrolyte interphase to prolong cycle stability of propylene carbonate-based batteries** — score: 0.400 The article discusses the use of alternative pulse current discharge techniques to enhance the solid electrolyte interphase (SEI) in propylene carbonate-based batteries, ultimately improving their cycle stability. The authors demonstrate that this approach significantly extends the lifespan of the batteries, suggesting potential advancements in energy storage technology. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Room-Temperature Operation of Thin-Film All-Solid-State Fluoride Shuttle Batteries Using CeF3 Anodes and Ag Cathodes** — score: 0.200 The article discusses the development of thin-film all-solid-state fluoride shuttle batteries that operate at room temperature, utilizing cerium trifluoride ($CeF3$) as the anode material and silver (Ag) as the cathode. These advancements could enhance the performance and efficiency of battery technologies in various applications. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Binary Additives Synergistically Modulate Crystallization for Enhanced HTL-Free Sn–Pb Perovskite Solar Cells** — score: 0.000 The article presents a novel dual additive approach using semicarbazide hydrochloride (SCH) and N-fluorobenzenesulfonimide (NFSI) to enhance the performance of hole-transport-layer (HTL)-free Sn–Pb perovskite solar cells. This strategy effectively controls crystallization kinetics, reduces defects, and stabilizes the material, leading to significant improvements in power conversion efficiency (22.73%) and open-circuit voltage (0.877 V), while also ensuring long-term stability. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Wave-Adaptive Triboelectric Nanogenerators Highly Compatible With Actual Wave Energy Spectrum for Autonomous Smart Ocean Platform** — score: 0.000 The article presents a wave-adaptive triboelectric nanogenerator (TENG) designed with a partial Ferris wheel structure, enabling it to effectively harness wave energy across a wide frequency range that closely aligns with real ocean wave spectra. This innovative design not only enhances the device's adaptability to varying wave conditions but also supports the development of an autonomous smart ocean platform capable of facilitating advanced marine sensing applications with long-distance data transmission. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **Non-Lattice Oxygen Triggered Deprotonation via Discontinuous Amorphous Interlayer on Supported IrOx for Acidic Oxygen Evolution** — score: 0.000 The study presents a novel catalytic design using a discontinuous WOx interlayer on TiN-supported Iridium catalysts for the acidic oxygen evolution reaction (OER). By incorporating isolated tungsten single atoms and amorphous WOx clusters, the catalyst shifts the reaction mechanism to a non-lattice oxygen-assisted deprotonation process, significantly improving catalytic activity and stability. The optimized Ir/W-TiN catalyst achieves impressive performance metrics, including high current densities at low overpotentials and exceptional durability, offering insights for future anode catalyst development. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **[ASAP] Enhanced Electrochemical Detection of Circularly Polarized Light Enabled by Spin-Selective Oxygen Evolution Reaction** — score: 0.000 The article presents a novel approach to improve the electrochemical detection of circularly polarized light through a spin-selective mechanism linked to the oxygen evolution reaction. This enhancement aims to increase sensitivity and specificity for light detection, potentially advancing applications in various fields such as photonics and optoelectronics. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Kinetic Scaling Rules Governing Phase Instability in Perovskite Halides** — score: 0.000 The article investigates kinetic scaling rules that influence phase stability in perovskite halides, a class of materials important for various applications, including photovoltaics. It emphasizes the role of kinetic factors in determining the stability of different phases, which is crucial for optimizing their performance in practical applications. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Chiral Helical Metal–Organic Frameworks with Intrinsic Spin Polarization Enable Enantioselective Electropolymerization** — score: 0.000 The article discusses the development of chiral helical metal-organic frameworks (MOFs) that exhibit intrinsic spin polarization, enhancing their capability for enantioselective electropolymerization. This advancement suggests significant potential for applications in chiral materials, potentially impacting fields such as drug development and organic electronics. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] In-Plane Electronic Metal–Support Interaction Enables Efficient Sulfur Catalysis on Ni Single-Atom Catalysts** — score: 0.000 The article discusses a novel approach to enhance sulfur catalysis using nickel single-atom catalysts, focusing on the importance of in-plane metal-support interactions. This mechanism is proposed to significantly improve the efficiency of these catalysts, offering insights into optimizing catalytic processes for various chemical reactions. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Cation Sampling Enables Regiodivergent Distal Functionalization of Ketones** — score: 0.000 The article reports on a novel cation sampling technique that facilitates the regiodivergent distal functionalization of ketones, allowing for the selective transformation of ketone substrates. This method enhances synthetic flexibility and enables the generation of diverse chemical products from a single ketone starting material. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Lead-Free Double Cs2Ag(Bi,Sb)(Br,I)6 Perovskites: Going below 1.8 eV Bandgap by Anion Exchange and Solid-State Reactions** — score: 0.000 This article discusses the development of lead-free double perovskites, specifically Cs2Ag(Bi,Sb)(Br,I)6, which achieve a remarkable reduction in bandgap to below 1.8 eV through anion exchange and solid-state reactions. The findings highlight potential advancements in the suitability of these materials for optoelectronic applications, offering a safer alternative to traditional lead-based perovskites. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Electroenzymatic Oxidative Desymmetrization by Engineered Thiamine-Dependent Enzymes for the Enantioselective Synthesis of Axially Chiral Biaryls** — score: 0.000 The article discusses the development of engineered thiamine-dependent enzymes that facilitate

electroenzymatic oxidative desymmetrization, enabling the enantioselective synthesis of axially chiral biaryls. This innovative approach highlights advancements in enzymatic catalysis for producing complex chiral compounds with potential applications in pharmaceuticals and materials science. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*

- **[ASAP] Iron Age: Ionic-Liquid-Mediated Interfacial Charge Transfer Enables Selective CO2 Photoreduction to Formic Acid on Iron Oxide** — score: 0.000 The article discusses a novel process for the selective photoreduction of CO2 to formic acid using iron oxide in conjunction with an ionic liquid, highlighting the role of interfacial charge transfer in enhancing this reaction. This approach presents a potential pathway for effective CO2 conversion, contributing to sustainable practices. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **[ASAP] Stimuli-Modulated Metal–Organic Framework (MOF) Reactivity toward a Three-Component Coupling Reaction** — score: 0.000 The article discusses the innovative approach of using stimuli-modulated metal-organic frameworks (MOFs) to enhance their reactivity in a three-component coupling reaction. By tuning the MOF's properties, the researchers demonstrate improved outcomes in chemical transformations, highlighting the potential for using such materials in advanced catalysis. Journal: *Journal of the American Chemical Society: Latest Articles (ACS Publications)*
- **Publisher Correction: Microbiota-induced T cell plasticity enables immune-mediated tumour control** — score: 0.000 The publisher correction addresses issues in the original publication related to the study on how microbiota influences T cell plasticity, which in turn plays a crucial role in immune-mediated tumor control. The correction aims to clarify findings and ensure accurate representation of the research's conclusions. Journal: *Nature*
- **Publisher Correction: GLP-1R–GIPR–PPAR / / quintuple agonism corrects obesity and diabetes in mice** — score: 0.000 The article presents a publisher correction regarding the findings on a quintuple agonist targeting GLP-1R, GIPR, and PPAR / / , which has been shown to effectively reverse obesity and diabetes in mouse models. The correction addresses previously reported details to enhance clarity and accuracy of the research results. Journal: *Nature*
- **Do you hate or love AI? Take Nature's poll** — score: 0.000 Nature is conducting a poll to gauge its readers' opinions on the increasing backlash against artificial intelligence and its effects on the field of science. The publication aims to understand whether readers harbor negative or positive sentiments toward AI's influence in scientific research and practice. Journal: *Nature*
- **A step-by-step guide for scientists who hate conference networking** — score: 0.000 The article by Caroline Dunne offers practical strategies for scientists who struggle with networking at conferences, aiming to alleviate anxiety and foster more effective interactions in professional settings. It provides a structured approach to navigating social environments in academia, ensuring that even those uncomfortable with networking can find ways to connect and build their professional relationships. Journal: *Nature*
- **Criminals are made, not born: how when you live shapes whether you will break the law** — score: 0.000 The article challenges the widely held belief that criminal behavior stems primarily from inherent character traits, instead emphasizing that environmental factors play a significant role in shaping whether individuals, particularly youth, are likely to engage in criminal activity. This perspective suggests a need to focus on social influences in preventing crime rather than solely attributing it to personal characteristics. Journal: *Nature*
- **Ebola outbreak is a global health emergency: what happens next** — score: 0.000 The article discusses the outbreak of the Bundibugyo virus, highlighting its status as only the third occurrence of this rare strain, which poses a global health emergency due to the absence of approved vaccines or treatments. It examines the potential implications and responses to the outbreak on an international scale. Journal: *Nature*
- **Can the 'steroid Olympics' show the sporting community how to support athletes better?**

— score: 0.000 The Enhanced Games, often dubbed the 'steroid Olympics,' highlight the existing shortcomings in how the sporting community supports athletes. By allowing performance-enhancing drugs, the event raises critical questions about fairness, health, and the ethics of competition in sports. Journal: *Nature*

- **Exclusive: Race begins to trial Ebola drugs amid current outbreak** — score: 0.000 Amid a current outbreak of the Ebola Bundibugyo virus, clinical trials for potential treatments are set to commence in the Democratic Republic of the Congo and Uganda, with researchers expressing confidence in the rapid mobilization of these efforts. Journal: *Nature*
- **Birds get a bad rap: why we should look up to our feathered friends** — score: 0.000 The article discusses the challenges facing various bird species due to threats to their survival, while highlighting the potential for positive change through conservation initiatives. It also emphasizes the remarkable intelligence of birds, suggesting that this aspect could play a crucial role in their recovery and the biodiversity of ecosystems. Journal: *Nature*
- **The Enhanced Games miss the point: science can clean up sport** — score: 0.000 The article critiques the concept of the Enhanced Games, arguing that promoting performance-enhancing substances could jeopardize athlete health and undermine the integrity of sports. It emphasizes the need for advancements in anti-doping science to effectively address these challenges. Journal: *Nature*
- **Daily briefing: Are we about to face a 'super' El Niño?** — score: 0.000 The article discusses the uncertain intensity of the impending El Niño weather pattern, highlighting that predictions are still evolving. Additionally, it points out discrepancies in lab-mouse strains which may not match existing classifications and explores innovative approaches in the search for new antibiotics amidst challenging circumstances. Journal: *Nature*
- **Daily briefing: Around seven hours of sleep slows biological ageing** — score: 0.000 The article discusses recent findings suggesting that getting between six to eight hours of sleep daily may significantly slow biological aging and reduce disease risk. Additionally, it highlights discoveries regarding interbreeding among ancient human species and the potentially detrimental effects of antibiotic appearances on antimicrobial resistance. Journal: *Nature*
- **An expandable cement-induced pseudo-supercapacitor brick for self-energy-storage buildings** — score: 0.000 The article presents a novel energy storage solution in the form of a full-cement pseudo-supercapacitor brick, which utilizes cement-based electrodes and magnesia-cement electrolytes. This innovative design allows the brick to serve not only as a structural building component but also as a self-energy storage unit, enhancing the functionality of energy-efficient buildings through integrated energy storage and photovoltaic capabilities. Journal: *RSC - Energy Environ. Sci. latest articles*
- **Formation and switching mechanism in high-performance perovskite memristors with ultra-wide dynamic range** — score: 0.000 The study presents high-performance perovskite memristors that function as efficient electrical switches by leveraging structural changes for switching. An innovative optical patterning technique enables precise control over the formation of switching sites, contributing to the memristors' ultra-wide dynamic range and minimal energy loss during operation. Journal: *Joule*
- **An Australian government scheme holds potential for supporting a just energy transition** — score: 0.000 The article discusses an Australian government initiative aimed at facilitating a fair energy transition, highlighting its potential to promote equity and accessibility in the shift towards sustainable energy sources. The scheme is positioned as a model for addressing socio-economic disparities while advancing environmental goals. Journal: *Nature Energy*
- **Hole-transfer cascade-engineered donor polymer for unencapsulated perovskite solar cells with improved moisture stability** — score: 0.000 The article presents a novel approach to improve the moisture stability of unencapsulated perovskite solar cells by utilizing a specially engineered electron-donating polymer with a deep highest occupied molecular orbital. This design by Min-Ho Lee

and colleagues aims to enhance the durability and performance of these solar cells in humid environments. Journal: *Nature Energy*

- **Layer photovoltaic effect in a two-dimensional antiferromagnet with parity–time symmetry** — score: 0.000 The article reports the discovery of a layer-dependent photovoltaic effect in bilayer CrSBr, a two-dimensional antiferromagnet exhibiting parity-time symmetry. The study highlights that the photocurrent generated can be manipulated through changes in the material's magnetic configuration, opening avenues for advanced optoelectronic applications. Journal: *Nature Materials*
- **Supramolecular dye polymers for aggregation-induced photocatalysis** — score: 0.000 The article discusses a breakthrough in photocatalysis through the use of supramolecular dye polymers, which maintain their reactive excited states even when aggregated. This approach leverages the rigidification of amphiphilic chromophores to facilitate effective light-driven production of hydrogen and hydrogen peroxide in aqueous environments, proposing a novel design principle for enhancing photocatalytic processes. Journal: *Nature Chemistry*
- **Defect-Rich Ceria with Ru Dopants Accelerating Surface-Mediated Disproportionation in Li-O2 Batteries** — score: 0.000 The article discusses the development of defect-rich ceria materials doped with ruthenium, which enhance the efficiency of surface-mediated disproportionation reactions in lithium-oxygen batteries. The findings suggest that these doped ceria materials could significantly improve battery performance by facilitating the necessary chemical reactions. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Chemomechanical Stability Determines the Performance of Dry-Processed Cathode and Separator Sheets in Thiophosphate-Based Solid-State Batteries** — score: 0.000 The article examines the role of chemomechanical stability in the effectiveness of dry-processed cathode and separator sheets used in thiophosphate-based solid-state batteries. It highlights how these materials' performance is influenced by their structural integrity under operational conditions, emphasizing the importance of material selection in battery design. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **[ASAP] Thermodynamic and Geochemical Simulations of Silicate–CO2–H2O Reactions for Carbon Mineralization** — score: 0.000 The study conducts thermodynamic and geochemical simulations to investigate the reactions between silicate minerals, carbon dioxide, and water, with a focus on understanding the processes involved in carbon mineralization. This research aims to enhance our knowledge of how these interactions can be harnessed for effective carbon capture and storage solutions. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] Integrating Molecular Simulations and Experiments to Explore Hydrogen Adsorption and Storage in Fluorous Metal–Organic Frameworks** — score: 0.000 The article discusses the integration of molecular simulations and experimental approaches to investigate hydrogen adsorption and storage capabilities in fluorous metal-organic frameworks (MOFs). It highlights the potential of these materials to enhance hydrogen storage efficiency, which is crucial for energy applications. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] An Interpretable and Streamlined Machine Learning Framework for Simultaneous Prediction of Polarization Curves and Maximum Power Density in Direct Methanol Fuel Cells** — score: 0.000 The article presents a novel machine learning framework designed for the simultaneous prediction of polarization curves and maximum power density in direct methanol fuel cells. This framework emphasizes interpretability and efficiency, aiming to enhance the performance and understanding of fuel cell operations. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] Time-Dependent Tensile Degradation and Associated Microstructural and Fracture Characteristics of Coal under ScCO2–H2O Conditions** — score: 0.000 The article investigates how coal's tensile strength deteriorates over time when exposed to supercritical carbon dioxide (scCO2) and water, focusing on the resulting microstructural changes and fracture behavior. The findings enhance the understanding of coal's performance in carbon capture and storage applications. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*

- **[ASAP] Catalytic Pyrolysis of Biomass Using Industrial Solid Waste: A Review and Outlook** — score: 0.000 The article reviews the process of catalytic pyrolysis of biomass utilizing industrial solid waste as a catalyst, highlighting recent advancements and potential applications in sustainable energy production. It discusses the benefits of this approach in enhancing biofuel yields and reducing waste, emphasizing the future outlook for incorporating waste materials into biomass conversion technologies. Journal: *Energy & Fuels: Latest Articles (ACS Publications)*
- **[ASAP] Simultaneous Enhancement of the Energy Storage Density and the Self-Healing Properties via Interface Grafting for Metallized Film Capacitor Application** — score: 0.000 The article discusses a novel approach to improve both energy storage density and self-healing capabilities in metallized film capacitors through interface grafting techniques. This advancement could significantly enhance the performance and durability of energy storage devices, paving the way for more efficient applications in various technologies. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **[ASAP] Morphology-Dependent -Fe2O3 Nanostructures as Bifunctional Electrocatalysts for Efficient Overall Water Splitting** — score: 0.000 The article explores -Fe2O3 nanostructures with varying morphologies, demonstrating their effectiveness as bifunctional electrocatalysts for both the hydrogen and oxygen evolution reactions in overall water splitting. The study highlights how these morphological variations enhance their catalytic performance, contributing to more efficient sustainable energy solutions. Journal: *ACS Applied Energy Materials: Latest Articles (ACS Publications)*
- **Atomic Scale Distortion Enables Optimal Cu–Co Coupling for Selective Electrochemical Nitrate to Ammonia Conversion in Neutral Electrolytes** — score: 0.000 The article discusses how atomic scale distortions in copper-cobalt (Cu–Co) catalysts enhance their efficiency in selectively converting nitrates to ammonia in neutral electrolytes. This findings suggest a promising avenue for optimizing electrochemical reactions, addressing environmental challenges related to nitrate pollution. Journal: *ScienceDirect Publication: Nano Energy*
- **China moves AI brain implants from trials towards real-world use** — score: 0.000 Chinese start-ups are advancing the development of brain-computer interface algorithms aimed at enhancing mobility and communication for individuals with disabilities. Following successful trials, these companies are focusing on bringing their AI-driven brain implants into real-world applications to improve users' quality of life. Journal: *Nature*